PART

Four

Defaultable Claims

CHAPTER 24

Credit Risk

Credit risk is the general term for the risk that a counterparty in a contract or security does not perform as required by the contractual obligation, leading to unexpected losses for the other counterparties or security holders. This type of risk must be distinguished from other types of risk, for instance that the value of a claim is less than expected due to some adverse development in the price of a given commodity. As will be discussed in Chapter 26, the distinction between the deterioration of the value of a claim and the ability of a counterparty to perform under a claim is very important. Some ABS will in many circumstances not fully repay without such failure to pay being an event of default.

24.1 DEFAULT, INSOLVENCY, AND BANKRUPTCY

Debt contracts, be it loans or bonds, usually bind the debtor in more ways than a simple payment obligation. It is therefore possible that a debtor violates the terms of a debt contract in ways other than not paying cash when due. This makes it important to distinguish between the various forms of of contract violations collectively usually listed as events of default in the respective contract.

For instance, the covenants of a loan or bond may restrict the gearing, or leverage, of a company. If the company exceeds the maximum permitted leverage, it is in default even though it may still be current on its payment obligations. How such a situation is resolved will depend on the debt contract or prospectus and may involve significant transfers of decision-making powers to creditors.

Generally, the concept of default is wider than that of insolvency. Insolvency refers to an inability to pay amounts when due, which may include payments for goods and services received, not only payments on money borrowed. Insolvency need not necessarily mean that a debtor has insufficient assets to cover its liabilities because some assets may be valuable, but hard to turn into cash in the time required to meet upcoming payments.

Bankruptcy is usually the final stage in a company's existence when it is insolvent and beyond rescue. For creditors, bankruptcy entails a number of disadvantages beyond delays in the recovery of money lent. Companies can have assets that only have value while the company operates, such as brand recognition and business relationships, that will be less valuable or even worthless in bankruptcy. This creates an incentive for creditors to work with a debtor to maximise the recovery value of a company's assets. In this sense, debt restructuring and debt forgiveness lead to losses for the creditors but such certain losses can be preferrable to the uncertainty of going through a court-administrated bankruptcy process.

24.2 SENIORITY AND SUBORDINATION

When a debtor is unable to repay monies borrowed, not all creditors have equal ranking in their claim to be paid. In essence, failure to pay means that there are insufficient assets to make payments and claimants to these payments will have to compete for the assets that can be distributed. According to their relative ranking, creditors are called senior or subordinated depending on the cardinal ranking of their claims. Relative seniority is a function of several factors.

24.2.1 Time subordination and acceleration

The most obvious form of subordination is time subordination: creditors who get paid first have a higher likelihood of recovering their money.

To prevent time subordination from overriding all other forms of creditor ranking, insolvency legislation in most countries ensures that all debts of a debtor normally become due and payable, including accrued interest, when a default occurs on a given class of debt. This acceleration protects holders of longer-term debt from time subordination, but can dramatically worsen the liquidity situation of the debtor. When a borrower uses multiple entities and governing laws to borrow, acceleration is usually made explicit through cross-default clauses in the relevant documentation. Such clauses reduce the risks related to situations where a debtor is insolvent in one jurisdiction, leading to a distribution of assets to debtors in that jurisdiction while holders of comparable debt elsewhere are time-subordinated.

While time subordination is undesirable in straight debt, it is being used to improve the credit tiering in some ABS transactions. In such structures, repayments of principal of the underlying loans is directed first to the most senior tranche so that this tranche has the shortest-weighted average life.

Some jurisdictions, e.g. Austria, limit time subordination in statutory terms by allowing for clawbacks of debt repayments in certain situations. In particular when a lender should be aware of the financial situation of a borrower and receives a payment shortly before bankruptcy, courts may order a return of such payments to the bankrupt estate for distribution to other creditors. The assumption behind such clawbacks is that the lender took advantage of superior information to extract early repayment to the disadvantage of less privileged creditors.

24.2.2 Contractual subordination

Private debtors can freely negotiate debt contracts between themselves, including a ranking of claims. In general, a debtor will offer a higher rate of interest on lower-ranking claims and a lender can choose to accept the additional risk inherent in a subordinated claim for this higher rate of interest.

Contractual subordination can take multiple forms. The most straightforward approach is to spell out seniority directly in the debt prospectus. However, other contractual clauses can have similar effects to explicit seniority/subordination specifications. For instance, certain classes of high-yield debt may offer the borrower the option to 'pay in kind' (PIK), i.e., pay with new debt instead of cash when coupons are due. This establishes a contractual time subordination of PIK bond holders because their claim to cash payment is in effect delayed while other debt holders may be receiving cash.

Some high-grade corporate bonds contain rating triggers on the coupons that increase the interest payable when the credit rating of the debtor falls below certain thresholds. Such rating triggers can increase stress in the free cashflow for the borrower just at the time when rating agencies are taking a negative view on the financial outlook of the debtor. Holders of bonds with rating triggers are affected to a lesser degree and are therefore somewhat senior to equivalent bonds without the rating trigger.

Lenders have a strong interest in their exact position in the seniority hierarchy of the borrower's debt. This implies that the hierarchy should stay largely unchanged over the lifetime of the bond. Covenants such as limits on leverage and limits to subordination of existing claims work to establish such stability.

24.2.3 Statutory subordination

Most countries' jurisdictions impose subordination on certain classes of creditors. The most prominent example is the statutory bail-in of certain bank creditors as part of the European Bank Resolution Directive and the TLAC[1]/MREL[2] requirements for bank funding.

Banks have traditionally relied on very low levels of equity, augmented by similarly low levels of subordinated debt, to finance their operations. This implied a very high share of senior debt, including deposits, in funding bank assets. When bank asset valuations declined drastically in the financial crisis that started in 2008, public authorities had the unpalatable choice of either accepting bank defaults on very large swathes of senior unsecured bank liabilities with unforeseeable consequences, or injecting additional equity through tax-funded bailouts. One of the results of this experience is that some form of bank debts can now be 'bailed-in', i.e., forced to share losses, even when there is no explicit contractual provision for this to occur.

Depending on the jurisdiction, the write-down features can be contractual (embedded in the bond prospectus) or statutory (enforced by national legislation across a class of debt instruments). It is not immediately obvious which of the two is preferrable. Contractual subordination allows banks to issue non-subordinated debt that is distinct from the subordinated debt. On the downside, investors are reminded by each prospectus that certain bonds are riskier than others and may require excessive compensation for

[1]Total loss-absorbing capital.

[2]Minimum Requirement for own funds and Eligible Liabilities.

this risk. Statutory subordination in effect moves the risk warning to legislation and so applies it equally across banks and a given instrument class.

A similar aim is achieved with conditionally convertible bank debt, also known as CoCo. CoCo bonds turn into equity automatically when certain capital thresholds are broken, usually related to the total tangible equity ratio relative to risk assets. CoCo conversion is therefore triggered by accounting events rather than supervisory action, and a CoCo holder receives equity rather than a given write down of her or his claims against the bank. The valuation of CoCos is complex, especially when one considers that the share price of a bank is likely to fall in a situation where capital is insufficient. Even if earnings expectations were unchanged, equity investors would consider dilution through CoCo conversion and potential additional equity issuance.

Statutory seniority is a much older concept, however. Most covered bond legislative frameworks protect certain counterparties of the cover pool more than other creditors of a covered bond issuer. Lenders to companies undergoing court-protected restructuring usually have certain protections ('lender in possession') and so on.

24.2.4 Joint liabilities and credit support

When debts are jointly owed by multiple debtors, the difference between joint and several liability and several liability is an important factor in default risk analysis. For several, but not joint, liability the default of any one borrower leads to a default on the debt[3]. The severity of this default (the loss given default LGD) may be low, however, given that only part of the debt is affected. For investors who apply historical cost accounting, such an event may trigger an impairment event, and hence force a re-marking of the position which may entail larger or smaller losses. Some rating agencies do not include LGD in these assessments and hence apply what is known as the weakest link approach: The credit rating of the joint debt instrument is in essence driven by the rating of the weakest borrower in the pool.

For joint and several liability, the debt only defaults if all borrowers default because the lost payments from any one borrower are covered by the others, up to the point where a single borrower is liable for the entire debt[4]. While this appears safer, it is often the case that no single borrower in the pool can in fact support the entire structure. Defaulting borrowers increase the burden on the non-defaulting borrowers, which in turn makes them more likely to default. Still, the credit risk of a joint and several liability is most affected by the risk of the strongest borrower in the pool.

In some cases, borrowers receive explicit or implicit debt support from a larger entity. For instance, Japanese regional entities are seen as benefiting from credit support by the central government or prefecture. This can result in a risk structure resembling a joint and several liability. An important question in such cases is the form of this support. The spectrum ranges from first-call unconditional guarantees, where the support provider makes an unconditional promise to immediately cover any missed payment from the debtor, to sufficiency support, where the support provider makes only a general

[3]In Boolean logic, this would be an OR relationship.
[4]In Boolean logic, this amounts to an AND relationship.

commitment to ensure the economic viability of the borrower. In such structures, debt holders may face delays on collecting due payments, or even the risk of reduced payments.

24.2.5 Sovereign debt

Sovereign debt differs fundamentally from private sector debt because the assets of a sovereign, such as taxes, levies and mineral rights, are only enforceable by the sovereign itself. This precludes the idea of liquidating a sovereign and distributing the assets to the creditors[5]. The sovereignty of a sovereign also precludes a court-directed asset distribution[6]. Put differently, what sets sovereigns apart from private borrowers is that they have a virtually infinite lifespan. While a private borrower can be wound down or dissolved when they fail to repay debts, sovereigns remain. This means that sovereign borrowing is a repeat game. Lenders to a sovereign will be aware of past defaults and accept debt on worse terms from repeat defaulters. Such experience is unavailable to lenders to private borrowers, and acts as a deterrent to capricious defaults by sovereigns.

In the course of the crisis in the euro area debt market, some solutions were proposed that rely on an effective mechanism to create senior and subordinated sovereign bonds. In practice, such ideas may clash with the very notion of sovereignty. Subordinaton of private debts is a fairly reliable concept because it can be pursued through the courts, and courts can also oversee the liquidation of private estates. Sovereigns, in contrast, have no or little distributable assets that can be liquidated by courts that are not also subject to legislation by the same sovereign. The freedom to decide which monies to pay to which creditor, meanwhile, is an essential ingredient of sovereignty.

Seniority in the absence of a court to enforce it is essentially only a promise by the issuer. Given that debt itself is a promise (namely, to repay money borrowed), senior sovereign debt is only more valuable than subordinated debt if the issuer breaks one promise (the promise to pay) but keeps another (to pay some creditors more than others).

24.3 THE DEFAULT PROCESS

The default of a debtor generally gives creditors the right to enforce their claims with a level of control that exceeds that present in performing lending relationships. The first

[5]It should be noted that this is the modern state of affairs. In history, taxation rights have been sold by sovereigns (cf. e.g. [41, 60]), and mining rights were granted in lieu of debt repayments [35].

[6]It should be noted that in modern democracies the executive branch of government is subject to the courts, but the sovereign itself is not. Courts can hold the executive to account when it fails to honour the law as it stands, but they have only limited jurisdiction about changes to the law through the legislative process. This process is only restricted by a country's constitution. A sovereign may also have bound itself through international treaties in a way that could restrict some commercial decisions. However, whether a government is able to bind the country in ways that restrict the application of its constitution is a perennial subject before the German constitutional court.

question for creditors is usually whether or not it is advantageous for them to allow the debtor to continue operating, possibly with some alterations to the business model, or alternatively wind down the debtor and realise the assets in a liquidation. The distinction between these two options is not always very sharp. For instance, a debtor may have a business that can be sold off in its entirety without suspending operations. In this case the operating business is comparable to an asset in its own right.

Under normal circumstances, asset sales are likely to incur substantial discounts to their carrying value. This is because production processes are usually very specialised and the default of a debtor is usually a sign that the specific approach of the debt was unviable. Potential buyers would need to integrate the assets into their production process, and may not have demand for the extra output that could be achieved through their use. Additional discounts are likely to arise in expedited sales processes ('firesales') when buyers perceive an opportunity to achieve lower prices because the seller can only choose between low valuation or no sale at all.

These factors often create a strong incentive for creditors to conduct the resolution of defaulted claims in a way that is non-antagonistic versus the debtor and involves a continuation of the debtor's business activities. The aim of such a process is an eventual recovery of the debtor to such a degree that liabilities can be serviced, potentially after reducing them by some amount. When this managed default approach is not feasible, a debtor might be wound down. Bankruptcy codes refect these different approaches by specifying multiple procedures. For instance, the US bankruptcy code[7] specifies a managed default (reorganisation) in Chapter 11 for private companies and Chapter 9 for municipalities, while a liquidation follows Chapter 7 of the that code for private enterprises.

Debtors usually have multiple debt instruments outstanding, and the preferred way to resolve a default depends for each investor on the specific nature of the instrument they hold. For instance, an investor who recently bought a low-coupon security well below par might prefer an immediate wind-down (resulting in an imminent cash payout perhaps in excess of the purchase price) while a holder of a high-coupon security would prefer a maturity extension and continuation of debt service rather than losing the coupon income.

In general, default proceedings have the potentially conflicting aims of respecting contractual priorities of different debt classes, and at the same time ensuring equitable treatment of debtors of equal rank. The specific quantification of 'equitable' alone is a challenge. To value claims of owners of securities with different coupons and maturities, one would have to resort to an appropriate discount curve. How to calibrate such a curve, and what valuations to use for such a calibration, can easily be subject to dispute.

The usual venue to resolve conflicts are the courts that have jurisdiction over the liabilities in question. However, the same debtor might have debt issued in multiple jurisdictions and hold assets that are distributed across yet a different set of jurisdictions. This creates the problem of multiple courts granting conflicting access to assets to different creditors. There is no obvious solution to this problem.

[7]Technically, Title 11 of the United States Code.

Related to the problem of multiple jurisdictions is the problem of hold-out creditors. Most debt holders are averse to holding illiquid securities such as debt instruments that are subject to default proceedings. They therefore tend to prefer a speedy resolution of default situations, even at the expense of maximising the recovery value. Sometimes this preference is abused by so-called hold-out investors[8] who delay the final resolution of a default through prolonged court actions with the aim of achieving a higher payout for themselves. In essence, a multiplicity of debt instruments and jurisdictions can create the preconditions for a Nash equilibrium that delays final resolution of claims in a manner that is equitable to all debt holders across all jurisdictions.

24.3.1 Collective action clauses

Collective action clauses, usually abbreviated to CAC, are an avenue to circumvent the problems related to court-directed debt restructurings. CAC are included in bond documentation and so form part of the legal framework of a debt security. A restructuring using CAC is a credit event as would be a restructuring in any other way.

Under a CAC, the issuer can propose certain modifications of debt instruments and bond holders can then vote on whether to accept those terms. The essence of the CAC process is that all bond holders agree to be bound by the outcome of this vote ahead of time by having invested in the bond. This differs from a court-directed default process where there is a risk that some creditors refuse a proposal from the issuer and try to achieve better terms in a court. Such court proceedings may hold up the default process and delay payments.

Multiple variations of CAC exist. The voting threshold for accepting modifications is usually high (generally 2/3 or 3/4 of nominal held), and various prescriptions for the determination of a quorum, and the procedure for dealing with the absence of a quorum exist. Some CAC require dual majorities when multiple bond series are to be restructured, namely majorities for both each affected series and across all series. Such cross-series modifications are a useful tool for ensuring equity between debt classes, and CAC requiring dual majorities, known as dual-limb CAC face a higher risk of failing. The alternative single limb CAC require only a majority across series.

It must be borne in mind that CAC merely govern the procedure to follow in a default situation to prevent adverse Nash equilibria from emerging. CAC do not cause defaults, and CAC do not cure defaults. If a CAC-based restructuring turns out to be unfeasible, and even improved offers from the debtor do not suffice to convince investors to agree, then a default will still go ahead. The way in which this happens will then resemble a court-ordered default with the potential delays that may arise if multiple courts rule on multiple debt instruments.

Euro area government bonds with original maturities longer than one year issued since January 2013 carry so-called model CACs due to a clause in the ESM Treaty. These dual-limb CACs were proposed by the Economic and Financial Committee of the EU and are included either in bond documentation, or the respective national law for bonds

[8] When investors acquire debt in near-default or default situations to then act as hold-outs, they are less charitably known as vultures.

issued without a specific prospectus under domestic fiscal legislation. The purpose of these clauses is to streamline any potentially required debt adjustment before the ESM engages in economic support for a euro area member state. The introduction of these CACs was an interesting practical experiment in having a coexistence of two types of otherwise comparable types of debt from the same issuer that only differ in the CACs. Market pricing normally does not distinguish between these debt types, which could be seen as a signal that CACs are of little relevance. There are two reasons to treat this conclusion with some caution, however. The first is that newer bonds are more liquid, which should attract a liquidity premium. This liquidity premium may offset any discount applied as a result of CAC. Second, and probably more significantly, it is not obvious that domestic law bonds without a CAC would be treated differently from CAC bonds if a restructuring were required. The issuer could in theory retro-fit CACs on bonds without such clauses through a change of domestic law, and this is what happened for the Greek ‘private sector involvement’ in 2012 [30]. A more educational experiment would therefore be the co-existence of CACs of different types.

24.3.2 Debt exchanges and consent solicitations

Not every case where a bond is not repaid as scheduled, or where bond terms are violated, is a default. Bond issuers can ask bond investors to accept changes in the terms of a bond, or to exchange their bonds, voluntarily. Such changes need not have a direct economic impact on bond holders, and may even serve to improve the probability of receiving the expected cash flows.

When an issuer needs to change certain terms of a bond, it can launch a consent solicitation where investors are asked to consent to this proposed change in terms. Usually in such cases, an issuer seeks to obtain a relaxation in some covenants that have turned out to be more onerous than expected. Investors can withhold consent, but may then face a higher risk of default. Because a consent solicitation modifies an existing instrument, the terms of the solicitation are usually prescribed, at least in general terms, in the bond prospectus.

An exchange offer, on the other hand, is a more comprehensive event because it involves a change in instrument. The issuer will typically propose an exchange of one bond into another, and offer some incentives to participate in this exchange. Typically, an issuer will offer an exchange of instruments with incentives paid to participating investors. Investors are free not to participate in the exchange, but those not taking part face two risks. The first is losing out on the incentives, the second is the reduced liquidity of the instruments that have not been exchanged if a large majority of investors agrees to the exchange.

Transactions of this kind are therefore prime examples of game theory. Investors have no a-priori interest to agree to any exchange that may leave them worse off in their claims versus the issuer. If investors as a whole resist the exchange, it will not succeed. The issuer, and its advisors, design the terms of the exchange such that a sufficient number of investors agree to it, and so in effect enforce the terms of the exchange on other investors.

An important consideration of events described in this section is that it is not correct to view them through the lens of an antagonistic relationship between investors and borrower. Investors rely on a borrower to use their funds in such a way as to earn them returns. The borrower is restricted in how these funds can be used by the terms of the debt instruments that the borrower has issued. When circumstances change, it may be acceptable to all parties to amend the terms of such instruments in such a way as to reflect the change in circumstances. The absence of antagonism is of course not the same as fraternity. Investors need to consider their own interests, and those of their fiduciaries, first.

Whether a bond exchange has an impact on a credit rating, or triggers credit default swaps, depends on the terms of the exchange. If it occurs under stressed conditions and results in a reduction of claims against the issuer, rating agencies may decide to treat the event as a default. Credit default swaps might also be triggered in this case. If the exchange does not reduce the economic value of the bond, there may be no impact on the rating.

24.3.3 Managed defaults

A managed default process starts with the decision not to pursue an immediate wind-down and requires a court decision to halt proceedings in this direction. The suspension of the wind-down is known as a stay. During the stay, debtors and creditors can conduct negotiations about potential debt restructuring, i.e., reduction or rescheduling of interest and principal payments. Debtors are also often able to incur new debts to cover working capital requirements[9]. Lenders advancing such funds are then subject to special protection (higher seniority). Ideally, a managed default results in a preservation and recovery of the business that the borrower engaged in, and creditors achieve the recovery of a higher share of their initial investment than would be coming out of a wind-down.

A managed default can fail if initial assumptions turn out to have been too optimistic. In such cases, a managed default will transition into a wind-down, a scenario discussed below.

24.3.4 Wind-downs

A wind-down is the last stage in the life of a corporate debtor. When there is no realistic prospect of a recovery in the entity's business, it may be preferrable to sell any remaining assets and pay out creditors with the proceeds. In rare cases, there may even be leftover funds that can go to erstwhile equity holders.

A wind-down is not necessarily a fast process and can take years. Banks in particular have assets that are contractual relationships which can span many years and are difficult to liquidate. The case of the German Herstatt Bank (which created the term Herstatt risk) is a good example. The wind-down of the bank started in 1974 and ended only in 2006.

[9]US law describes this situation as 'debtor in possession'.

24.4 CREDIT RATINGS

Given the difficulty faced by investors when trying to assess the credit risk of a given debt instrument, it is unsurprising that an industry has developed that provides credit risk analysis services. These companies, generally called rating agencies[10], issue opinions and express relative credit risks in the form of credit ratings. The ratings from any one agency form a scale, and examples of comparable long-term ratings from five major rating agencies are given in Table 24.1. Similar ratings, but with simpler scales, exist for short-term debt instruments. The best possible rating is AAA/Aaa while issues in default are rated C or D. While ratings are generally public, the reports that support and explain these ratings usually require additional access fees. Ratings above the BBB-/Baa3 level are usually called investment grade and this threshold is reflected in the inclusion criteria for investment-grade debt indices. Debt with lower ratings is called variously 'sub-investment grade', 'high yield', 'speculative' or 'junk'.

Ratings can apply to issuers, classes of debt by one issuer, or individual issues. Issuer ratings generally have one additional category, namely 'selective default' which describes a situation where the issuer has chosen not to service some debts but is still current on others.

Rating agencies update ratings either on a regular schedule or in response to incoming data. Such rating changes, at least those undertaken by the leading agencies (the three leftmost agencies in Table 24.1) are generally watched closely by the market. As part of this process, agencies sometimes given an indication that a rating change is imminent through such qualifiers as 'review for upgrade' or 'negative outlook'. Markets react to the publication of such qualifiers and to their removal just as they react to changes in ratings themselves. Actions by the smaller agencies attract less attention in international markets but are still relevant. DBRS is recognised by the Eurosystem General Framework [6] and therefore relevant for euro area collateral eligibility while R&I is the leading agency opining on Japanese domestic issuers.

Credit ratings generally refer to the likelihood of full and timely payment of principal and the full payment of interest[11]. Some rating agencies also take into account the likely loss in case of a default event. Significantly, credit rating scales are ordinal, but not cardinal, scales. Higher ratings imply a lower likelihood of loss than lower ratings, but no rating implies a given level of credit risk. This means that credit ratings should not form an input to quantitative risk models although they invariably are used in this way. For instance, Article 59 of the Eurosystems General Framework uses one-year probabilities of default to translate external ratings[12] into Eurosystem quality steps.

Every rating agency publishes a number of reports outlining its rating methodology for different types of instrument. Usually, these start with a stand-alone issuer rating

[10]Some people take issue with the use of the word 'agency' in this context because this term is usually reserved for public institutions, not private companies. As will be explained later, of all the concerns surrounding rating agencies, this one is probably the least significant and the term 'agencies' is used here throughout.

[11]Note that timeliness of payment of interest is not part of this definition.

[12]A rating agency is called external credit assessment institution (ECAI) in this framework.

TABLE 24.1 Comparable long-term debt rating scales of five major credit rating agencies (Standard and Poors, Moody's, Fitch, DBRS and Ratings & Investment). The horizontal line below BBB- delineates the investment grade from lower ratings.

S&P	Moody's	Fitch	DBRS	R&I
AAA	Aaa	AAA	AAA	AAA
AA+	Aa1	AA+	AA (high)	AA+
AA	Aa2	AA	AA	AA
AA-	Aa3	AA-	AA (low)	AA-
A+	A1	A+	A (high)	A+
A	A2	A	A	A
A-	A3	A-	A (low)	A-
BBB+	Baa1	BBB+	BBB (high)	BBB+
BBB	Baa2	BBB	BBB	BBB
BBB-	Baa3	BBB-	BBB (low)	BBB-
BB+	Ba1	BB+	BB (high)	BB+
BB	Ba2	BB	BB	BB
BB-	Ba3	BB-	BB (low)	BB-
B+	B1	B+	B (high)	B+
B	B2	B	B	B
B-	B3	B-	B (low)	B-
CCC+	Caa1	C	CCC (high)	CCC+
CCC	Caa2	C	CCC	CCC
CCC-	Caa3	C	CCC (low)	CCC-
CC/C	Ca	C/DDD	CC/C	CC
D	C	DD/D	D	D

(which need not be on a scale comparable to the final rating scale) to which upward and downward adjustments are made to reflect external support or unfavourable environments. For each rated instrument or instrument class, further adjustments are made to reflect rank in the liability structure of the issuer, as well as presence, quantity and quality of any collateral. In many cases, a sovereign ceiling is then applied so that no debt instrument of a domestic issuer can have a higher rating than the sovereign in which it is incorporated.

A credit rating is normally issued on request, and against payment from, the issuing entity. Such a solicited rating creates a conflict of interest between the agency's economic interests and its objectivity. To manage this conflict, agencies generally opine only on concrete structures rather than advising on what steps an issuer would have to take to achieve a given target rating. This service of ratings advisory is instead performed by investment banks. There is as a result no process of negotiation of rating versus payment between issuer and agency. On the one hand, this limits the scope for ratings of convenience, but on the other hand also protects the agencies from responsibility for issuer structures that later turn out not to be stable. Because some ratings, particularly those of banks and sub-sovereign entities, depend on the ratings of other entities, namely those of their sovereigns, agencies can also issue an unsolicited rating in order to have this required input. Sovereign ratings of developed economies tend to be unsolicited because these issuers have no need to pay for a credit rating to achieve investor participation. Although an unsolicited rating has less scope for conflicts of interest, rating agencies have usually privileged access to financial information of the issuers with solicited ratings. The absence of that additional information could negatively affect the quality of unsolicited ratings. Of course, for developed economy sovereigns there is a plethora of publicly available information far in excess of what would be available for a typical bank, for instance. As a result, there is no unequivocal difference in quality between solicited and solicited ratings. Some regulators, however, insist on a clear delineation between the two types of rating.

Issuers can choose to discontinue their ratings from a given agency. This of course may be perceived as taking evasive action ahead of a looming downgrade, and, to the extent that ratings are required by some investors, may lead to a loss of potential demand. On the other hand, an issuer may have a substantiated difference of opinion with a rating agency over methodology and may use this to justify such a step.

The perhaps more serious concern of an issuer-funded ratings system is not that an individual issuer obtains favourable ratings in exchange for cash, but that entire debt classes are rated inconsistently relative to other debt classes. This is because a ratings agency may forgo the income from a single transaction to protect its reputation but may be more conflicted when having to choose between assigning non-viable ratings to a whole class of debt, or securing a larger income stream. Some of the ratings issued for structured products ahead of the financial crisis in 2008 were based on rather simple models; one could argue that a more critical stance from the agencies would have prevented some of the excesses in leverage at the time. However, one should not neglect the co-dependency of issuers and professional investors who typically act as agents, not principals, in the investment process. As long as fund managers are incentivised to achieve certain returns with a given credit rating, they have limited interest in setting aside overly optimistic assessments from a rating agency, at least not in a structural way.

24.4.1 Rating migration

Credit ratings present an apparent contradiction: it is clear that longer-dated debts have a higher probability to default than shorter-dated debt (because there is more time for them to default in), but ratings are assigned uniformly to all comparable liabilities of the issuer. This contradiction is resolved by rating migration. Credit ratings reflect the information available at the time they are assigned but it is understood that they can change. While a longer-dated bond has the same rating as a shorter-dated bond by the same issuer, this rating can change, possibly for the worse, over time, including after the shorter-dated bond has matured.

The migration of ratings is an additional reason to treat ratings as ordinal rather than cardinal rankings. Lower rated securities can be less risky than higher rated ones,

TABLE 24.2 Global one-year corporate average transition rates (1981–2017, in %) published by Standard and Poors (from Table 21 in [83], transposed). NR stands for Not Rated. Source: Data from Diane Vazza and Nick W Kraemer. 2017 annual global corporate default study and rating transitions. Technical report, S&P Global Ratings, 2018.

To/From	AAA	AA	A	BBB	BB	B	CCC/CC
AAA	86.99	0.51	0.03	0.01	0.01	0	0
AA	9.12	86.95	1.72	0.1	0.03	0.02	0
A	0.53	7.91	88	3.45	0.12	0.08	0.12
BBB	0.05	0.5	5.22	85.79	4.88	0.18	0.21
BB	0.08	0.05	0.3	3.73	77.19	5.05	0.59
B	0.03	0.07	0.12	0.49	6.79	74.34	13.18
CCC/CC	0.05	0.02	0.02	0.11	0.58	4.44	43.46
D	0	0.02	0.06	0.17	0.68	3.59	26.82
NR	3.15	3.97	4.52	6.16	9.72	12.28	15.63

but an assessment of absolute risk involves more characteristics than just a rating. Ratings are generally meant to apply 'through the cycle', i.e., be independent of the current state of the business cycle. However, because business cycles are not fully predictable, rating transitions can have cyclical components. That being said, investors also have only partial insight into business cycle dynamics. The informational value of ratings is therefore not diminished by the potential errors in business cycle assessments by the rating agencies.

Rating agencies generally publish statistics about rating changes (cf. e.g. [66, 83, 85]) which serve to underpin the reliability of their ratings. These statistics generally include annual default rates as well as annualised rating change probabilities. An example is shown in Table 24.2.

Rating migration means that care needs to be taken in interpreting these statistics. For instance, a given rating agency may show that a AA-rated security has a one-year default probability of $p_{AA,D}$, meaning that a security with this rating will default within the year with a probability of $p_{AA,D}$. The default probability over two years $p_{2,AA,D}$ could therefore assumed to be given by the sum of the probabilities of defaulting in the first year and surviving the first year but defaulting in the second, i.e.,

$$p_{2,AA,D} = p_{AA,D} + (1 - p_{AA,D})p_{AA,D} \approx 2p_{AA,D} \tag{24.1}$$

This interpretation would be incorrect because it neglects the possibility of surviving the first year, but having a different rating at the end of it, and then defaulting with the probability implied by the new rating. The correct approach would therefore be:

$$p_{2,AA,D} = p_{AA,D} + (1 - p_{AA,D}) \sum_k p_{AA,k} p_{k,D} \tag{24.2}$$

where k runs a sum over all non-default ratings that the security can have after the first year. Comparing this to the previous equation shows that the two are identical if, and only if, $p_{AA,k} = \delta_{AA,k}$, i.e., if the rating cannot change.

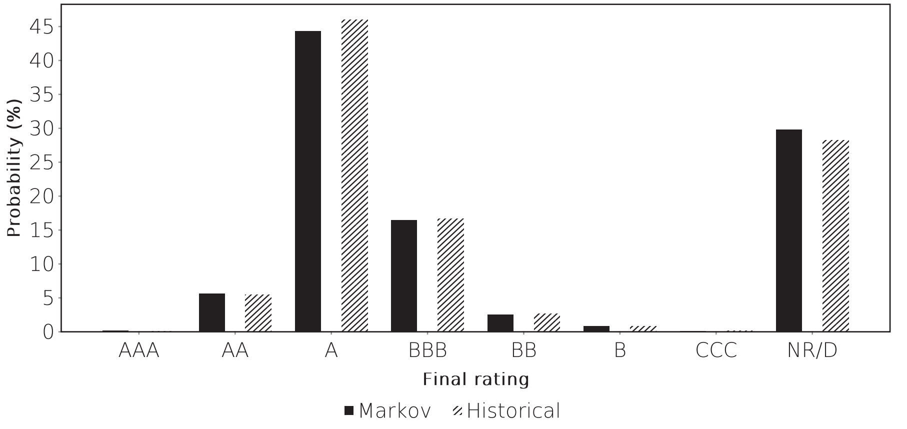

FIGURE 24.1 Comparison between the seven-year Markov-implied and actual probability distribution of the rating of an A-rated issuer after 7 years. Data from SP RatingsDirect [83].

The most basic assumption on credit ratings is that they follow a Markov process determined by a transition matrix as implied by Table 24.2. This approach underpins the Jarrow–Lando–Turnbull credit model [73]. The raw transition matrix is augmented by an additional column that makes the default states (D and NR in this classification scheme) absorbing. Multi-year ratings transition probabilities are then given by the powers of this one-year transition matrix. The Markov assumption amounts to the postulate that the transition process has no memory and is homogeneous over time. In practice, there are reasons to assume that this postulate may be violated. The business cycle can drive unforeseen changes in ratings, and companies actively manage their credit rating through changes in their capitalisation. A company may choose to increase gearing by distributing capital through dividends or share buybacks, or reduce gearing through retaining earnings or raising additional equity. The Miller–Modigliani theorem that the gearing of a company is irrelevant is not only contradicted by the tax effect on debt financing [46], but also because there is a non-linear relationship between debt financing cost and credit ratings. While an issuer may be forgoing tax benefits by keeping gearing low enough to maintain a very strong rating, increasing gearing may lead to such a low credit rating that the borrower would be required to pay very high coupons, or post collateral for borrowing and bilateral derivative transactions. This would suggest that the rating of a single company may exhibit mean-reverting behaviour.

It is possible to test the Markov assumption because rating agencies publish not only one-year transition probabilities but also track ratings transitions over longer periods. If the ratings process were non-Markovian, a comparison between the Markov-implied multi-year transition matrices and the actual multi-year transitions would show a difference.

Figure 24.1 shows a simple example of such a calculation. The one-year transition matrix derived from Table 24.2 is used to construct a seven-year transition matrix which can be compared directly with the actual seven-year transition matrix produced by the

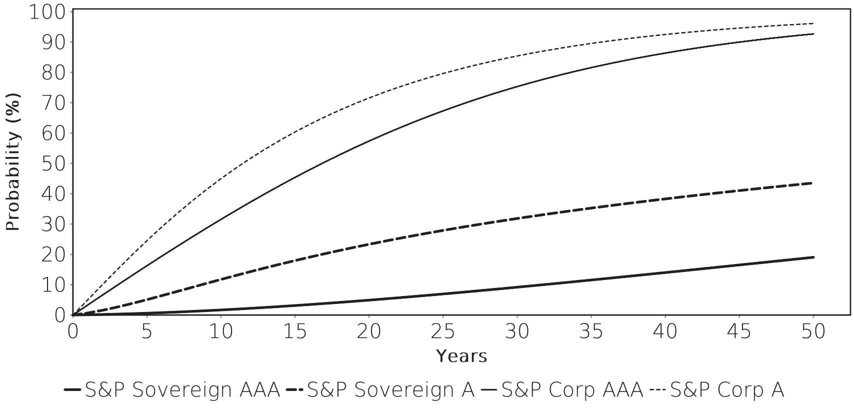

FIGURE 24.2 Cumulative probability of losing an investment grade rating over time for two different initial ratings and issuer types. Sources: Data from Diane Vazza and Nick W Kraemer. 2017 annual global corporate default study and rating transitions. Technical report, S&P Global Ratings, 2018; Lawrence R Witte. 2017 annual sovereign default study and rating transitions. Technical report, S&P Global Ratings, 2018.

same rating agency for the same issuer types. Shown in this example are the probabilities of an initially A-rated issuer to find itself 7 years later in any rating band under both approaches. Although some probabilities differ slightly, overall the Markov assumption does not perform too badly in this example[13].

The no-memory property of the Markov process implies that the initial rating is irrelevant for the credit risk of an issuer over very long horizons. Simply speaking, over a long enough time, any rating will migrate to any other rating with sufficient probability. This is unless the issuer gets trapped in the absorbing default state from which there is no return.

For a sovereign issuer, the default state is not absorbing because, as outlined above, it cannot be wound down. A rating agency may be tempted to conclude that sovereign ratings should reflect the possibility of strategic defaults. On the other hand, sovereigns may be more inclined to preserve good ratings given that there is no tax effect rewarding gearing and investment by domestic companies may be constrained by a low sovereign ceiling. An interesting comparison is therefore between the transition matrices of sovereign and corporate issuers, as done in Figure 24.2. The projected 50 year risk evolution of the corporate issuers shows an appreciable convergence at long investment horizons, i.e., the expected irrelevance of the initial rating. For sovereign issuers, this convergence is not reached in the time horizon shown here. Note that transition to Not Rated status is a large contributor to the significantly higher probability of losing investment grade ratings for corporate issuers (cf. the last row in Table 24.2) while sovereigns tend to preserve at least unsolicited ratings. For an investor, the loss of an investment grade rating may lead to forced selling but the associated losses may be

[13]Note that because the observation horizon is finite, there is a small sample bias between the historical 1-year and 7-year transition matrices.

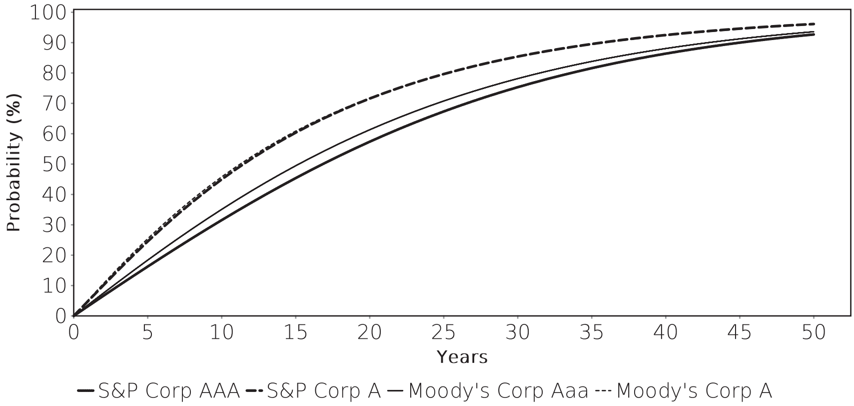

FIGURE 24.3 Comparison of cumulative probabilities of losing an investment grade rating (same definition as in Table 24.2) for two different initial ratings and two different rating agencies. Sources: Based on Sharon Ou, Sumair Irfan, Yang Liu, and Kumar Kanthan. Annual default study: Corporate default and recovery rates, 1920–2016. Technical report, Moody's Investor Services, 2018; Diane Vazza and Nick W Kraemer. 2017 annual global corporate default study and rating transitions. Technical report, S&P Global Ratings, 2018.

much less than in the case of a default. It is also notable that the difference between sovereign and corporate issuers is much larger than the difference between AAA and A ratings.

At the same time, one can compare the ratings evolutions across different rating agencies, as in Figure 24.3. Note that this comparison is not entirely meaningful because the historical rating transitions refer to distinct (albeit largely overlapping) pools of obligors, there is a slight difference in the observation period, and the two agencies may use slightly different definitions for a given rating. On this evidence, the risk characteristics of equal ratings assigned by the two agencies shown here are not materially different. While there is some divergence in the AAA rating evolution, the two A ratings follow a virtually indistinguishable path. This can be interpreted as a sign that ratings do reflect all available information in a rational way. It also supports the judicious use of ratings for the quantification of credit risk. Detractors might instead argue that ratings are not sufficiently independent. This will be discussed below.

24.4.2 Alternative rating approaches

The credit rating business is a natural oligopoly with high barriers to entry. For a new entrant to obtain the attention and trust of investors it is not sufficient that it is correct in its credit assessments, however defined. It would also require existing agencies to be wrong to such an extent as to make investors look for alternatives. To demonstrate superiority, a new pretender would have to demonstrate over a sufficient number of defaults (which translates into a sufficiently long period of time) that its own ratings were better predictors of default risks than those of the existing providers. To conduct the necessary analysis, it would have to rely on funding and data from issuers for this extended period of time. Additionally, regulators have placed additional burdens on rating agencies that require costly compliance mechanisms to implement.

In the European Union, a feeling that the major rating agencies, which tend to be incorporated outside the EU, had been unduly harsh in their assessment of European issuers, led to calls for new approaches to credit risk assessment. The idea that a domestic agency might take different views on domestic issuers is not completely absurd. The Japanese R&I rating of the Japanese sovereign is AAA, far above the single-A ratings assigned by Western rating agencies. Fitch was also the last of the three major agencies to strip France of its AAA rating at a time when it was the last of the three major agencies to still have a large European ownership interest.

Two potential alternatives to the current setup have been discussed. The first is to set up rating agencies that are funded by investors or the public sector, the second is to reduce the usage of ratings in financial markets.

The analysis services provided by rating agencies are valuable, which exposes them to the curse of the commons: Every potential user values the product but not every user is willing or able to pay for the service. The unwillingness of some users to pay for the product reduces the incentives of those who are able to pay. Issuers are paying for this part of market infrastructure because it supports their own funding levels, leading to conflicts of interest. If the users of ratings wanted their own ratings service, they could have set it up a long time ago.

To break the curse of the commons, one could introduce publicly funded rating agencies. The problem with that approach is that the public sector itself may be perceived as conflicted. Few governments are expected by investors to encourage a publicly funded company to opine objectively on that government, regional entities or significant banks under its jurisdiction, if such opinions were to require negative judgement. If the problem of 'he who pays the piper calls the tune' exists for issuer-funded ratings, it might well be that 'he who pays and regulates the piper calls the tune'. Taxpayers might also view ratings as too narrow a common good to warrant public funding.

Alternatively, the use of agency ratings could be discouraged. This is the course of action taken by the European Union which is trying to regulate greater reliance on internal ratings analysis by banks and asset managers. Aside from avoiding any biases in the analysis of agencies, this would reduce the impact of correlated selling or buying decisions that can result from rating changes by a single rating provider when it is being used to define index memberships or collateral eligibility. The essential problem with this solution is cost. There is not much added value in analysing the same public accounting information $n + 1$ times when it has already been analysed n times by competent analysts, or, worse, by assigning a less than competent analyst to conduct additional analysis that then influences investments. To date, therefore, while over-reliance on agency ratings is being discouraged, these ratings still form a very important part of the credit investment process.